BM32 · XIA MERCURY SDD · XIA_DET0

SDD Calibration Across Beamtimes

Does the energy calibration and detector response measured on the AXO C10 standard in March 2026 apply to the InPoSi map of September and the NIO295 map of October? Each experiment is calibrated on its own fluorescence lines, then every calibration is tried on every dataset.

Prepared 2026-10-19 · pyXRT branch sdd-calibration (fit_calibration, fit_air_path) · fits on map-averaged spectra (maps) or summed counts (C10)

Summary

ANSWER	No. The March C10 measurement does not transfer. Its calibration places the September and October peaks **6–11 eV too high** and makes their fits worse than the old default (χ^2 ×3.4 on InPoSi, ×1.6 on NIO295). Its resolution (242 eV FWHM at 5.9 keV) and air path (15 cm) do not match either.
SETUP	The March and September/October data were taken with different detector settings. The September and October files record the XIA configuration `mercury_0p5musec_AXAS_3_10eVpBin_2048Bins.ini` (0.5 µs peaking time) and give 144–156 eV at 5.9 keV. The March file records no configuration; its 242 eV resolution points to a shorter peaking time.
SEP ⇔ OCT	September and October agree with each other to **1–2 eV** between 3 and 12.6 keV. Fitting NIO295 with the InPoSi calibration costs 0.6 % in χ^2, and the reverse 3.5 %.
OLD DEFAULT	The old default (10.032 eV/ch, −31.3 eV) is within ±4 eV of the September/October calibrations between 9 and 13 keV (Ga, As, Ir, Pb lines), but **13 eV low at In Lα** (3.3 keV) and 10 eV low at Nd Lα.
RECONFIG	In September the detector was reconfigured during the reference measurement: the first two Ge counts sit **31 eV** away from the third, taken 3 minutes later. The third agrees with the InPoSi map calibration to 0.1 eV. A calibration is only valid after the last configuration change.
IN-MAP	Peak positions are stable during the maps (< 2 eV over 2.5–4 h), but the resolution is not: InPoSi shows a **step** of the Ga K width from 208 to 226 eV after ~50 min, and NIO295 a **slow rise** from 151 to 160 eV over 4 h.
AIR PATH	From line ratios: 15 ± 3 cm in March, 26 ± 1 cm on InPoSi in September (with or without the layer model). NIO295 gives 26–40 cm depending on the excitation model, because self-absorption in the bulk crystal dominates its line ratios. The notebooks currently assume 20 cm.

The three datasets

Dataset	Proposal	Date	Measurement	Fitted range	Elements in the fit	Layer model
C10	blc16918	2026-03-03	ct 100 s	2.5–19.5 keV	Pb La Pd Mo Cu Fe + Ca Ar	none (thin film)
InPoSi_v2_center	ihma907	2026-09-10	81 × 81 map, 1 s	2.9–12.6 keV	In Ga As Cu Ni Ca Ar	InP 50 nm / InGaAs 1.9 µm / Si
NIO295_laue_map_precise	ihhc4366	2026-10-12	91 × 91 map, 1 s	4.3–15.8 keV	Nd Ir Pt Pb	$Nd_2Ir_2O_7$ bulk

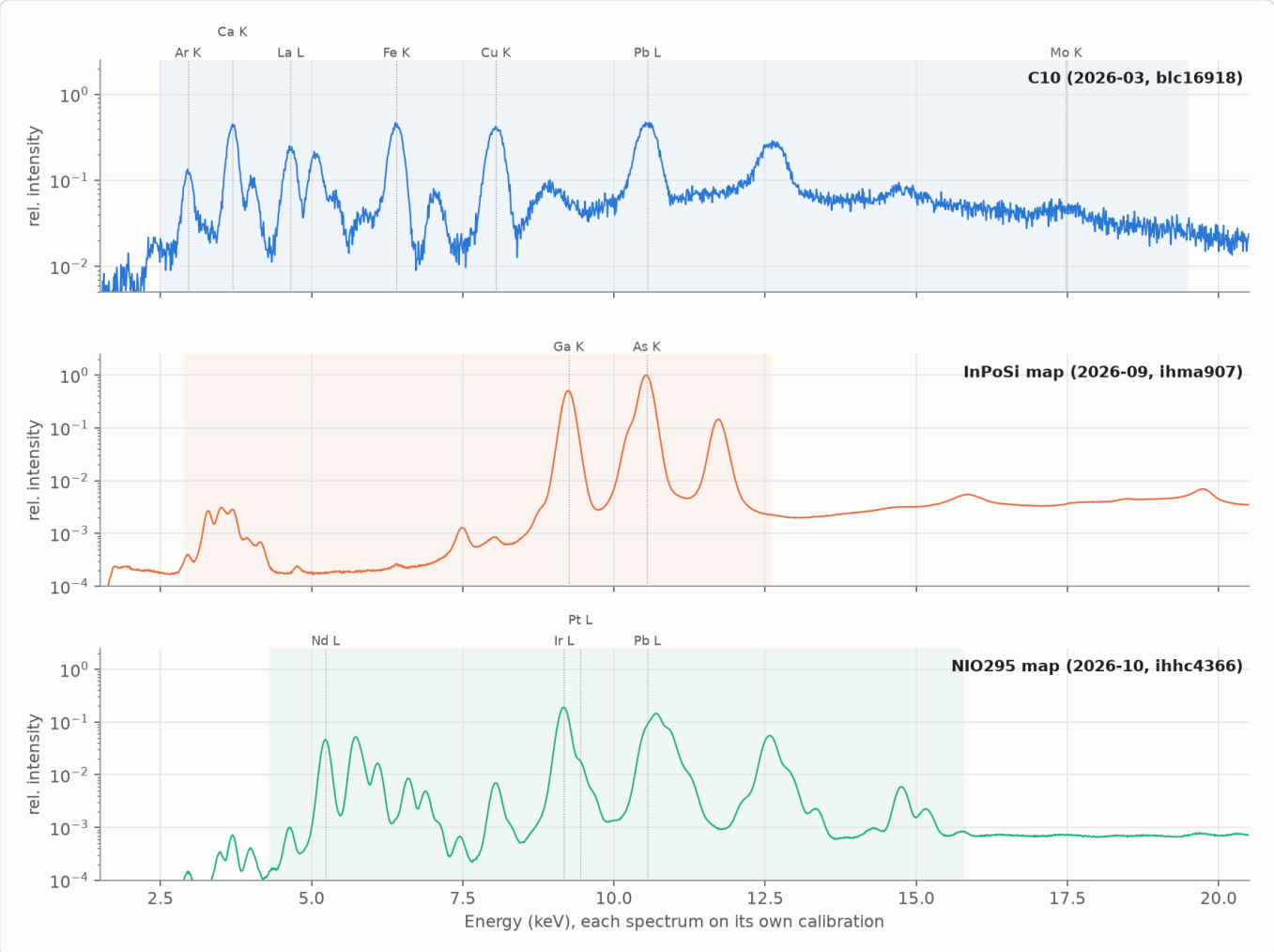


Figure 1. The spectra used, each on its own calibration. Shaded: fitted range. Labels mark the reference line of each element group. The C10 spectrum has broader peaks than the two maps, which is the resolution difference discussed below.

Each experiment calibrated on its own lines

Gain, offset, resolution (noise and Fano terms) and sample–detector air path are fitted together, with every line at its tabulated energy and tabulated relative intensity. The maps include your layer models, since self-absorption changes

the line ratios inside each group. On NIO295 that matters for the calibration itself: without the bulk model the calibrated energies move by up to 6 eV across the fitted range.

Dataset	Gain (eV/ch)	Offset (eV)	FWHM 5.9 keV	FWHM 10 keV	Air path	Reduced χ^2
C10 · Mar	10.01889 ± 0.00168	-7.09 ± 1.15	242	315	15.2 ± 0.7 cm	1.903
InPoSi · Sep	10.01702 ± 0.00006	-13.81 ± 0.06	156	196	25.7 ± 0.0 cm	1238
NIO295 · Oct	10.01341 ± 0.00005	-11.15 ± 0.05	144	177	31.3 ± 0.0 cm	784.5
old default	10.03200	−31.26	135 (assumed)	–	20 (assumed)	–

> The errors on the map fits are statistical only. With 6 500–8 300 pixels averaged, the counting errors are tiny and the reduced χ^2 (~800–1200) is set by model imperfections, so the real uncertainties are much larger than quoted: a few eV for energies, a few cm for the air path.

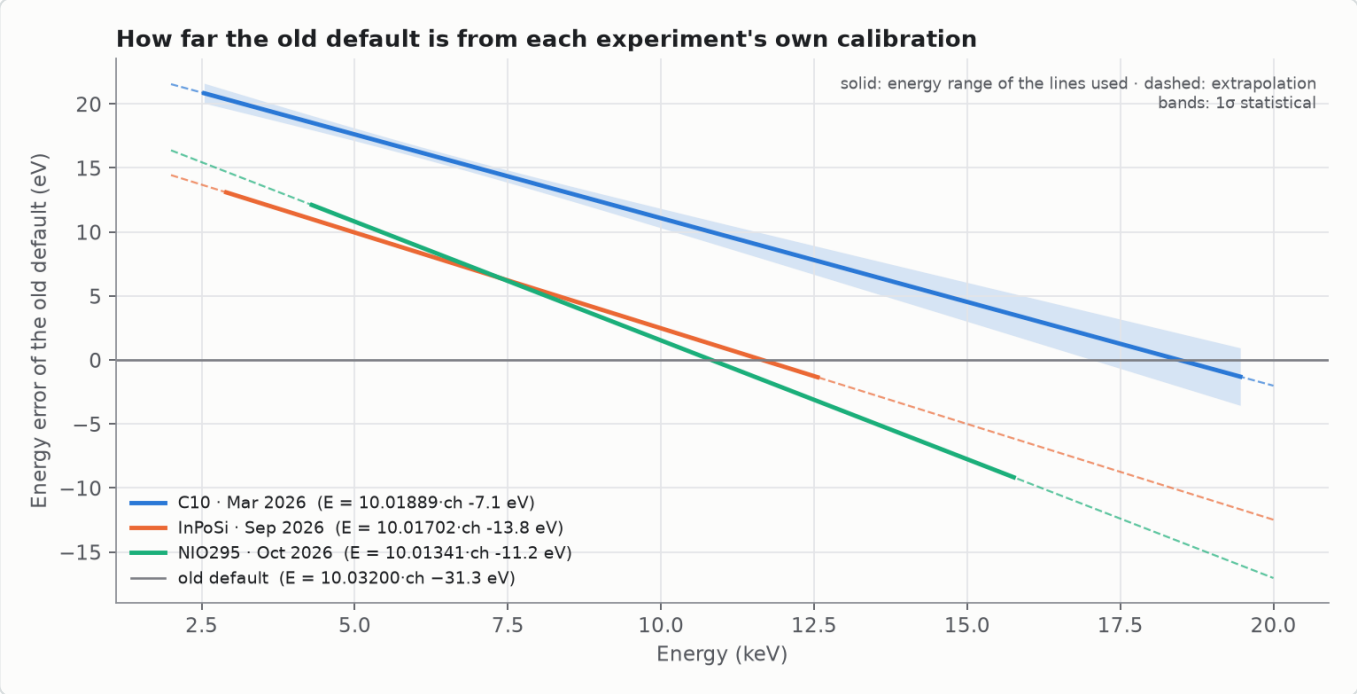


Figure 2. What the old default gets wrong in each experiment: the energy it assigns to a peak minus the true energy, using each experiment's own calibration as the truth. September and October nearly coincide; March sits 6–9 eV above them.

Energy error at the lines you analyse

For a peak recorded in a given experiment, the error in eV if a different calibration is used to read it. Zero on the diagonal by construction.

Line	Energy	Recorded in	old default	C10 (Mar)	InPoSi (Sep)	NIO295 (Oct)
In Lα1	3.287 keV	InPoSi · Sep	-12.5	+7.3	0	+1.5
Nd Lα1	5.230 keV	NIO295 · Oct	-10.4	+6.9	-0.8	0
Fe Kα1	6.404 keV	C10 · Mar	-15.8	0	-7.9	-7.6
Ir Lα1	9.175 keV	NIO295 · Oct	-3.1	+9.1	+0.7	0
Ga Kα1	9.252 keV	InPoSi · Sep	-3.6	+8.4	0	-0.7
As Kα1	10.544 keV	InPoSi · Sep	-1.7	+8.7	0	-1.1
Pb Lα1	10.552 keV	C10 · Mar	-10.4	0	-8.7	-9.8
Pb Lβ1	12.614 keV	NIO295 · Oct	+3.3	+11.0	+1.9	0

Every calibration on every dataset

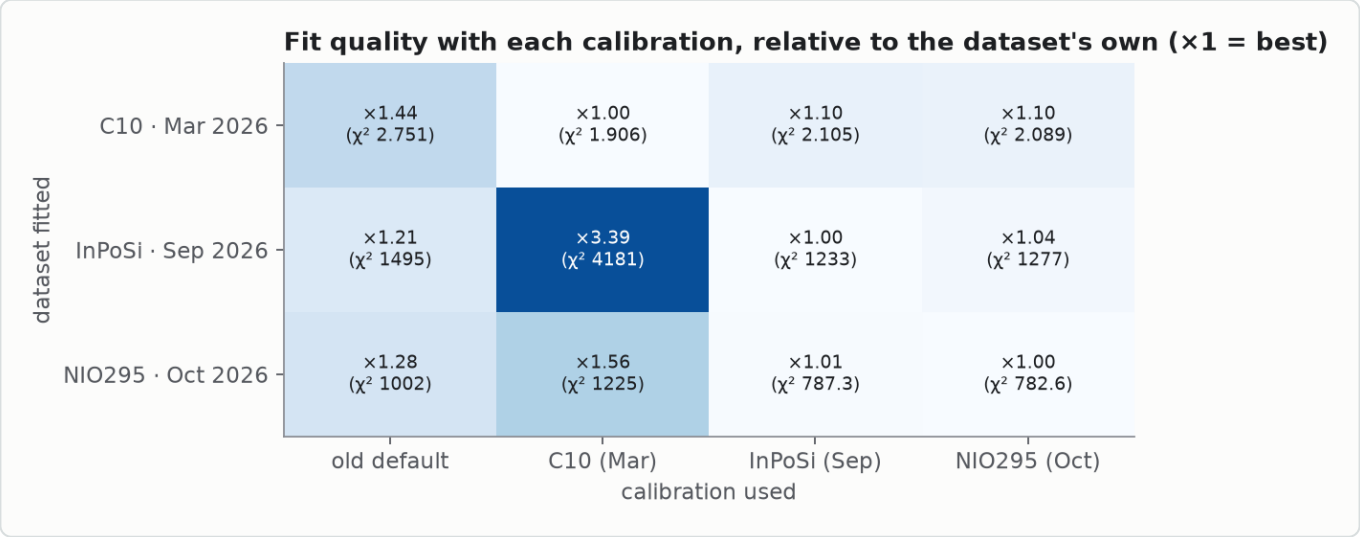


Figure 3. Reduced χ² of each dataset fitted with each calibration held fixed (resolution and air path still free), divided by the χ² with its own calibration. The C10 calibration is the worst choice for both maps, worse than the old default.

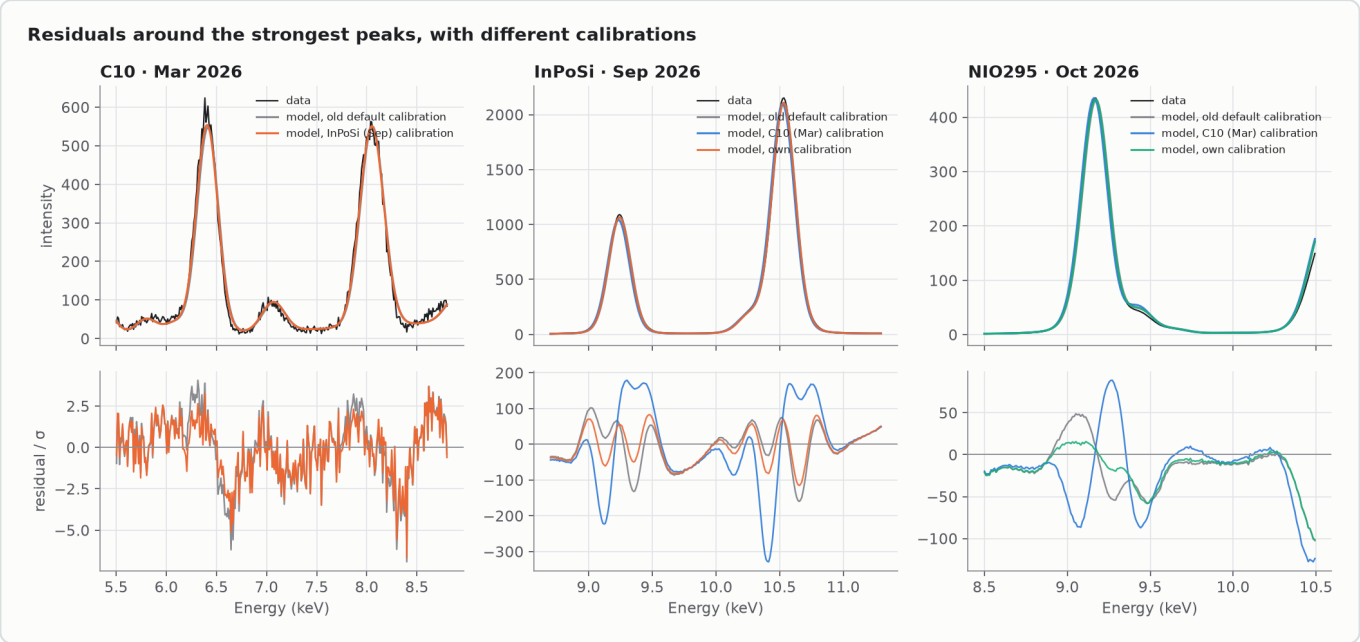


Figure 4. Where the misfit comes from. With the C10 calibration (blue) the maps show the derivative-shaped residuals of a position error on every strong peak. On the C10 data, the September calibration is only slightly worse than its own, because the C10 peaks are broad and the counting statistics are low.

Peak by peak

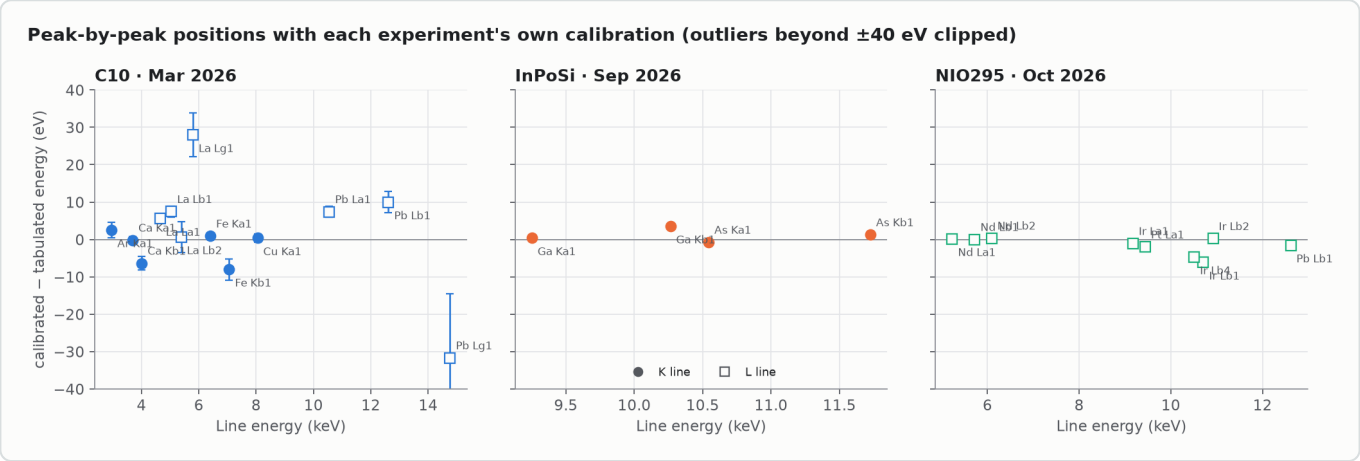


Figure 5. Each strong peak refitted alone with only an energy shift free, with each experiment's own calibration. Filled circles are K lines, open squares L lines. In March the L lines of La and Pb sit 6–10 eV above the K lines. In October the Nd and Ir L lines scatter within −6 to 0 eV. The InPoSi range only has K lines strong enough for this test (Ga, As).

> The K/L difference seen in March cannot be tested in September (no strong L line) or October (no K line). If it is real line-shape physics it should appear in every experiment, so a C10 measurement with the 0.5 µs configuration would settle it.

Resolution

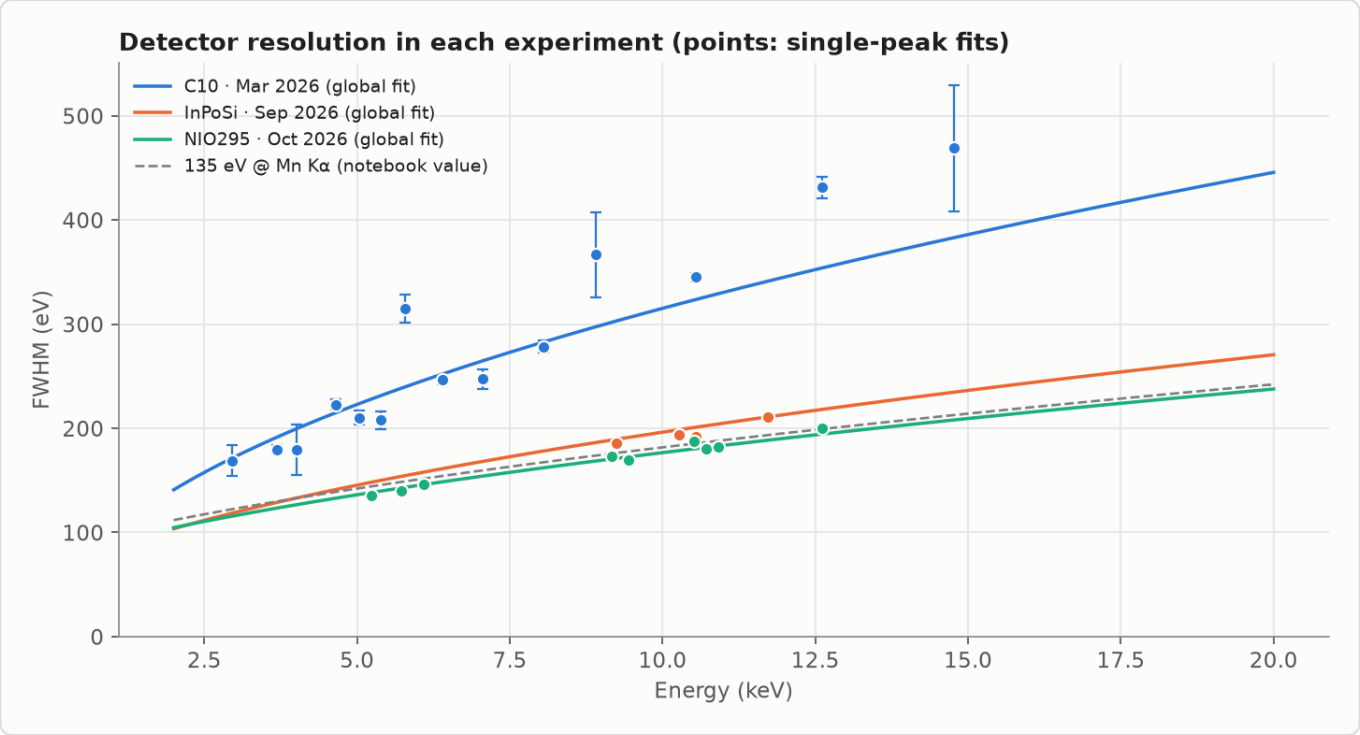


Figure 6. FWHM against energy from the global fits (lines) and from single-peak fits (points). September and October sit close to the 135 eV-at-Mn-Kα curve the notebooks start from (within ~15 %); March is 55–80 % broader, consistent with a different peaking time. The scatter of the March points above its curve at 9–15 keV comes from the Pb L blends.

Stability within a beamtime

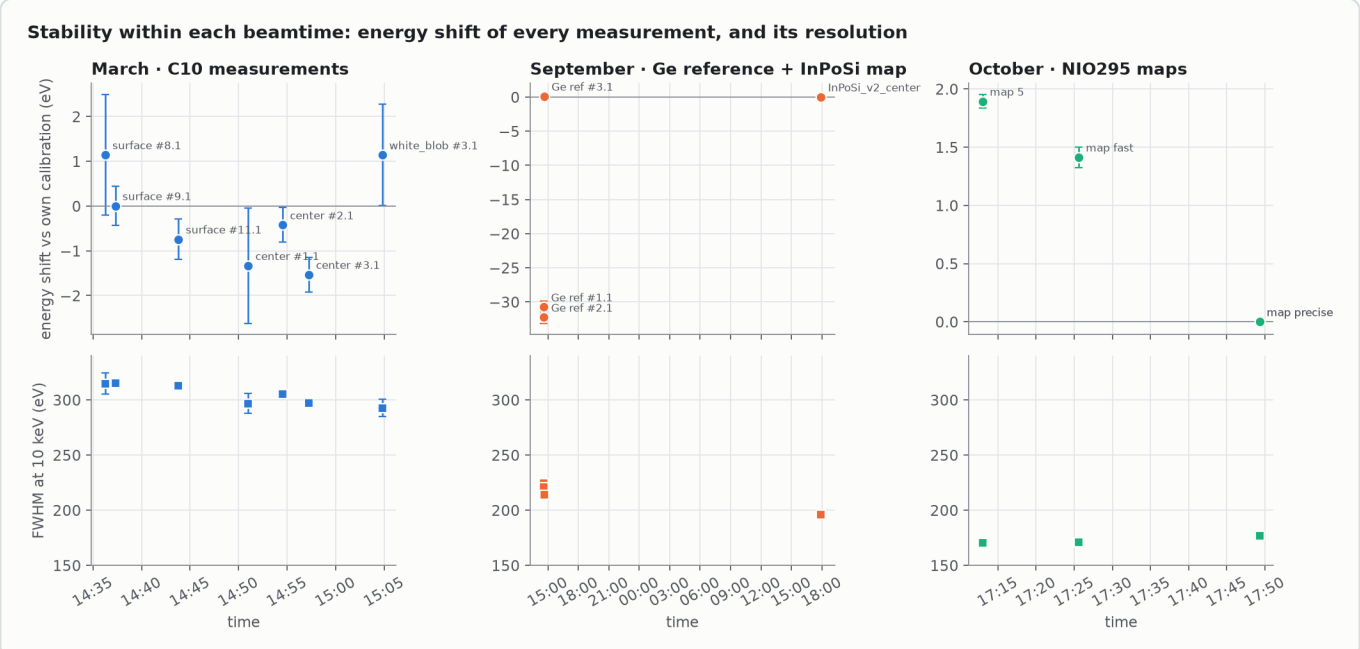


Figure 7. Each measurement fitted with its beamtime's gain fixed and only an energy shift free; width evaluated at 10 keV. March: seven measurements over 30 min agree within ±1.5 eV. September: only the Ge reference file has XIA spectra besides the map; its first two counts are 31 eV off, the third matches the map to 0.1 eV. October: three maps over 40 min agree within 2 eV.

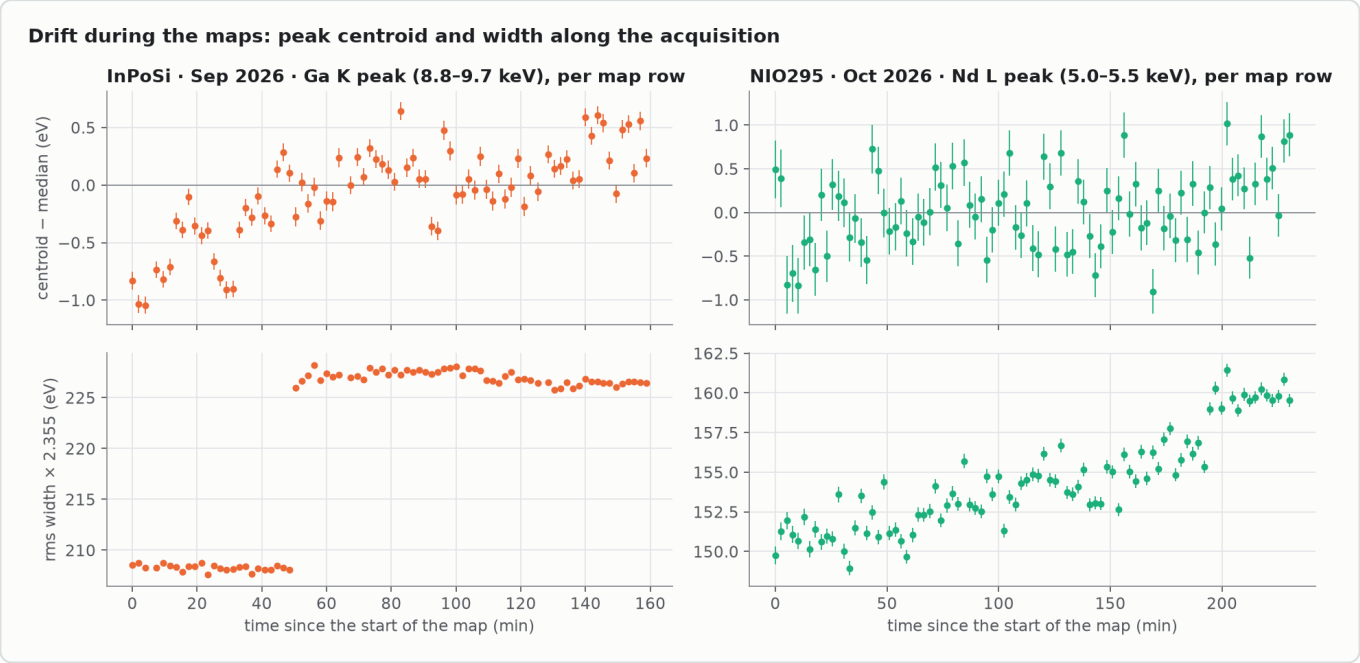


Figure 8. Inside the maps, one point per row of 81 or 91 pixels. The centroids stay within 1.7 eV (InPoSi) and 1.9 eV (NIO295). The widths do not: InPoSi steps up by 9 % after about 50 min, and NIO295 rises steadily by 6 % over 4 h. Your notebooks already have a segment and linear-width option for this; the plots show it is needed on both maps.

Sample–detector air path

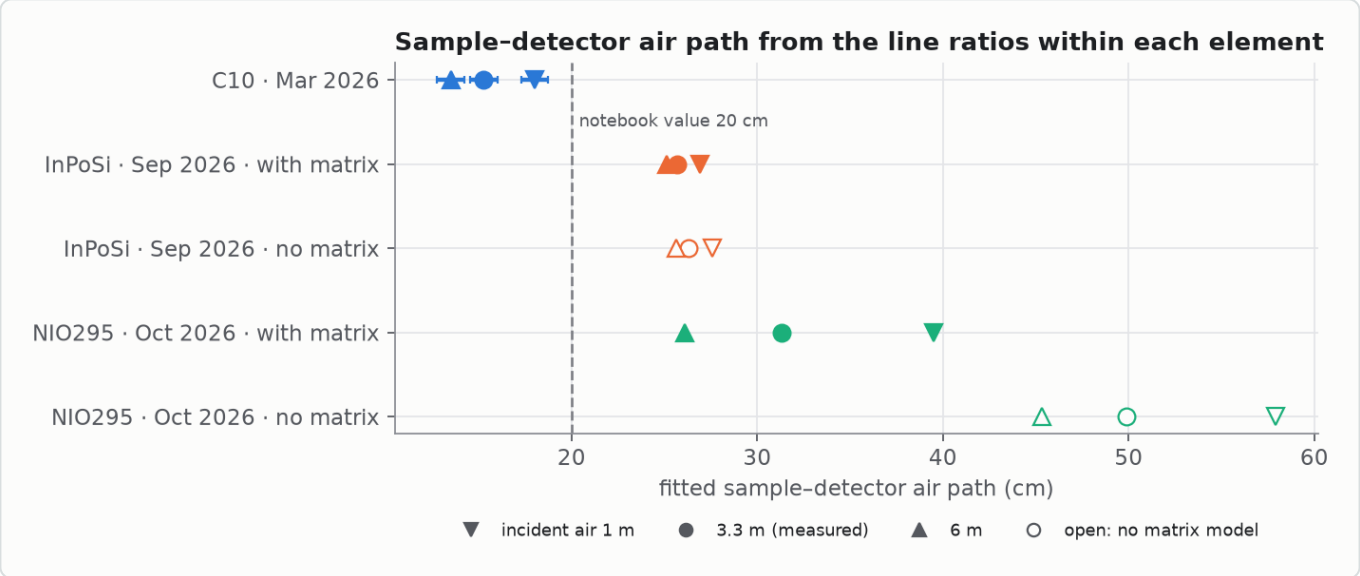


Figure 9. The air path fitted from the line ratios within each element, for three incident air paths (1, 3.3 and 6 m, the excitation model's main uncertainty) and with or without the layer model. C10 and InPoSi are robust to both. NIO295 is not: in a bulk crystal self-absorption shapes the line ratios, and the result moves from 26 to 58 cm with the model choices.

> 15 cm in March against 26 cm in September suggests the detector was mounted at a different distance. Combined with the different resolution, the March measurement looks like a different detector setup altogether.

What this means for the analysis notebooks

- **Calibrate per beamtime, after the last detector configuration change.** Calibrated energies changed by 6–11 eV between March and September, and by 31 eV within minutes when the configuration was loaded.

- **A sample's own lines are a usable calibration when no standard was measured.** Both maps self-calibrate to well within the resolution, and September and October agree to 1–2 eV. The limitation is energy coverage: InPoSi constrains only 3–12.6 keV, NIO295 4.3–15.8 keV.
- **Measure the C10 standard once with the 0.5 µs configuration.** It would give a calibration over 2.5–19.5 keV for the current setup, test whether the K/L offset is real, and give the air path on a thin film at the current detector position.
- **Keep resolution free and segmented.** The resolution changes inside the maps, so a single detector resolution is not enough even within one scan.
- **Air path:** 26 cm from InPoSi is the best current estimate for the September/October position; NIO295 cannot pin it down.

How this was computed

- Excitation: BM32 white beam built from the slits, mirrors and ring current recorded in each scan (`bm32_from_h5`), incident air path 3.3 m. Detection efficiency: 450 µm Si, 1 µm C window.
- Fits: constrained Voigt model of `xrf_fit` with escape peaks, polynomial background (order 5, 3 and 2 as in the notebooks), `fit_calibration` and `fit_air_path`. Layer models with primary fluorescence only (secondary fluorescence off for speed).
- Map spectra: monitor- and live-time-normalised average over all pixels, counting errors propagated. C10: summed raw counts with Poisson errors.
- Single-peak checks: each strong line refitted in a ±1.5 FWHM window with the global response fixed and only an energy shift (or the resolution) free.
- In-map drift: `peak_width_profile` on the cleanest strong peak chosen by `width_window`, one bin per map row.